JAMES CATHERINE RICHARDSON

Research Engineer

Email: james.richardson@gmail.com

Phone: +44 7821 456392

LinkedIn: linkedin.com/in/jamesrichardson42

PROFESSIONAL SUMMARY

I'm a Research Engineer with seven years of experience building scalable systems and developing novel solutions to complex technical problems. I've spent the last four years focusing on machine learning infrastructure and distributed computing, working across academia and industry. I genuinely enjoy mentoring junior engineers and translating research findings into production-ready code. Based in Glasgow, I'm looking for roles where I can continue to push technical boundaries while contributing to impactful projects.

WORK EXPERIENCE

Senior Research Engineer

Quantum Analytics Ltd

Glasgow, United Kingdom

www.quantumanalytics.io

March 2022 - Present

Designed and implemented a distributed machine learning pipeline processing 500GB of data daily, reducing inference latency by 45% through custom CUDA kernel optimization.

Led a team of three junior engineers in developing feature extraction modules for time-series analysis, establishing code review standards that improved system reliability by 60%.

Conducted research into federated learning approaches for privacy-preserving model training, publishing findings in two peer-reviewed conferences and filing one provisional patent.

Collaborated with product teams to transition experimental research into production systems, managing full lifecycle from prototyping through deployment to 2 million active users.

Research Engineer

Catalyst Systems

Edinburgh, United Kingdom

www.catalystsystems.co

September 2019 - February 2022

Architected and deployed containerized microservices using Kubernetes and Docker, improving system throughput by 35% while reducing infrastructure costs by 22%.

Developed Python libraries for GPU-accelerated data processing, which became the foundation for three subsequent product releases and reduced processing time from hours to minutes.

Investigated novel approaches to anomaly detection in sensor networks, implementing algorithms that improved detection accuracy from 78% to 94% across test datasets.

Presented technical findings at four industry conferences and contributed to internal knowledge sharing through twelve technical workshops delivered to 80+ colleagues.

Software Engineer

DataFlow Innovations

Glasgow, United Kingdom

www.dataflow-innovations.net

June 2017 - August 2019

Built RESTful APIs and backend services in Python and Go, supporting core platform functionality used by 150+ enterprise clients.

Optimized database queries and implemented caching strategies using Redis, reducing average response times from 800ms to 120ms and improving user satisfaction scores by 18%.

Participated in on-call rotation and maintained 99.7% system uptime across production infrastructure hosting critical financial data processing pipelines.

Mentored two graduate trainees through their first year, establishing mentorship practices later adopted across the engineering team.

EDUCATION

Doctor of Philosophy in Computer Science

University of Edinburgh

Edinburgh, United Kingdom

2014 - 2017

Thesis focused on distributed algorithms and fault tolerance in edge computing environments.

Bachelor of Science in Computer Science with Mathematics

University of Glasgow

Glasgow, United Kingdom

2010 - 2014

First Class Honours. Specialization in algorithms and systems programming.

TECHNICAL SKILLS

Programming Languages: Python, Go, C++, Java, SQL, Bash

Data Technologies: Apache Spark, TensorFlow, PyTorch, Pandas, NumPy, Kafka

Infrastructure: Kubernetes, Docker, AWS, GCP, Terraform, Jenkins

Databases: PostgreSQL, MongoDB, Redis, Cassandra

Machine Learning: Deep Learning, Federated Learning, Time Series Analysis, Feature Engineering

Other: Git, Linux, distributed systems, system design, technical documentation

CERTIFICATIONS AND AWARDS

AWS Certified Solutions Architect Professional, 2021

Best Technical Innovation Award, Catalyst Systems, 2021

Certified Kubernetes Application Developer, 2020